

Research Paper Assistant

An AI-Powered RAG Assistant for Summarizing and Querying Research Papers

Project Report

Submitted by: Taniya Choudhary

B.Tech Computer Science (AI)

Indraprastha Engineering College (AKTU)

September 2026

1. Problem Statement

Reading and understanding academic research papers is time-consuming, especially when a student or researcher needs to quickly grasp the core idea, contributions, and limitations of a paper before deciding whether to read it in full. Manually skimming a 10-15 page paper to extract this information is inefficient and often leads to missed details.

The Research Paper Assistant addresses this by allowing a user to upload a research paper (PDF) and instantly receive a structured summary, key contributions, limitations, and future work, along with the ability to ask free-form questions about the paper's content. This reduces the time needed to evaluate a paper from tens of minutes to under a minute.

2. Architecture

The system follows a Retrieval-Augmented Generation (RAG) pipeline, which grounds the language model's responses in the actual content of the uploaded paper rather than relying on the model's general knowledge. This ensures answers are accurate and specific to the document provided.

2.1 High-Level Flow

User → RAG → LLM → Response

The flow works as follows: the user uploads a PDF research paper through a simple web interface. The document is parsed and split into smaller text chunks to fit within model context limits. Each chunk is converted into a vector embedding using Google's Gemini embedding model and stored in a FAISS vector index. When the user requests a summary or asks a question, the query is embedded the same way, and the most relevant chunks are retrieved from FAISS based on similarity. These retrieved chunks, along with the user's request, are passed to the Gemini language model, which generates a grounded, context-aware response.

2.2 Components

- Frontend: Streamlit web app for PDF upload, action buttons (Summary, Key Contributions, Limitations, Future Work), and a chat box for free-form questions.
- Document Loader: Extracts raw text from the uploaded PDF.
- Chunking: Splits the paper into overlapping text segments to preserve context across boundaries.
- Embedding Model: Google Gemini embeddings convert each chunk and each query into vector representations.
- Vector Store: FAISS stores and retrieves the most relevant chunks based on cosine similarity.
- LLM (Gemini): Generates the final summary, structured insights, or answer using the retrieved context.

3. Prompt Design

Four fixed prompt templates were designed for the core actions, each instructing Gemini to answer strictly using the retrieved context so responses stay grounded in the paper rather than the model's general knowledge:

- Summary: "Using only the provided context from the paper, write a concise summary covering the paper's purpose, method, and main result."

- Key Contributions: "List the key contributions of this paper as bullet points, based only on the provided context."
- Limitations: "Identify the limitations or weaknesses the authors mention or that are evident from the provided context."
- Future Work: "Extract any future work or research directions mentioned in the provided context."

A fifth, open-ended prompt template supports the free-form Q&A feature, instructing the model to answer the user's question using retrieved chunks and to state clearly when the paper does not contain enough information to answer, in order to reduce hallucination.

4. Evaluation

The assistant was tested by uploading a small set of sample research papers and manually verifying the outputs against a human reading of the same papers.

4.1 Method

- Uploaded 2-3 research papers of varying length and domain.
- Compared the generated summary, contributions, and limitations against a manual read of each paper.
- Asked several free-form questions with answers known in advance to check factual accuracy.

4.2 Observations

- Summaries and key contributions closely matched the paper's actual content when the paper had a clear structure (abstract, sections, conclusion).
- Retrieval quality depended on chunk size — very small chunks sometimes split key sentences, slightly reducing answer completeness.
- The model correctly declined to answer when a question was outside the scope of the uploaded paper, showing the grounding was working as intended.

5. Challenges

- Chunking large PDFs while preserving enough context for accurate retrieval without exceeding model input limits.
- Ensuring extracted PDF text stayed clean (some PDFs had irregular formatting, headers/footers, or multi-column layouts that affected text extraction quality).
- Balancing retrieval count (how many chunks to pass to the LLM) against response speed and cost.
- Preventing hallucination — tuning prompts so Gemini answers only from retrieved context instead of filling gaps with general knowledge.

6. Future Enhancements

- Add citation highlighting so answers link back to the exact section/page of the paper they came from.
- Support multiple file formats beyond PDF, such as DOCX or arXiv links.
- Add a persistent history so users can revisit previously analyzed papers without re-uploading.
- Improve chunking with semantic/section-aware splitting instead of fixed-size splitting.